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SUBJECT OF THE INVENTION

[**0001**] The present invention relates to a radon Tron measurement device with improved detection sensitivity of the charge nuclide.

[**0002**] Background of the Invention is a radioactive isotope radon (222Rn: atomic number 86) and Tron (220Tn: atomic number 86) is present as a gas in the air. For this reason it is necessary to pay attention to the body exposure by inhalation.

[**0003**] In recent years, radon-Tron as environmental radiation that is emitted from the concrete wall has been a problem, radiation handling facilities there has been a growing need for measurement of radon-Tron also in well general of the building of the room within or basement, etc. .

[**0004**] In FIG. 5, radon decay chain (A) and Tron of decay series (B) are shown. Only the left edge of the radon and thoron is a gas, and their daughter is a solid. As will be described later, nuclide their daughters excluding radon and thoron is that the α-emitting or β-emitting and emits α rays are positively charged.

[**0005**] Figure 4 is an example of a conventional radon thoron measuring device is shown. In the figure, the sample gas (measurement gas) is first sent to a dust filter 10 is a large foreign matters such as dust are removed. Next, the ion pre-recipe stator 12, ions contained in the sample gas by using the electric field is removed. This, in the next ion chamber 14, ionization charge due to unwanted ions is provided in order to prevent it from being the collector measured. As a result, it has become a result of daughter nuclides of radon daughter nuclides and Tron is also removed.

[**0006**] In the ionization chamber 14, by α-rays radon and thoron is released at the time of the collapse, the gas in the ionization chamber 14 is ionized, this time resulting ionization charges are trapped in the center electrode 14a by the action of an electric field. The output signal from the ion chamber 14 caused by this, via a vibrating capacitor electrometer head 16 is sent to a vibration capacitance electrometer 18, thereby being displayed concentration of radon thoron. Incidentally, the flow meter 20 and the pump 22 are connected to the ion chamber 14.

[**0007**] The present invention is, however, above-mentioned conventional apparatus, it was impossible to measure the radon and thoron separately. In the above conventional apparatus, since the radon

daughters and thoron daughters are removed by ion pre recipe theta 12, it could not be determined for those nuclides. Incidentally, it is desirable to improve the measurement sensitivity of the radon-Tron.

[**0008**] The present invention has been made in view of the above conventional problems, and an object thereof is respectively separately measure radon and thoron.

[**0009**] Further, the present invention aims to realize a measurement of the radon daughters and thoron daughters.

[**0010**] Furthermore, the present invention aims to improve the measurement sensitivity.

[**0011**] The invention of claim 1, wherein in order to achieve the above object, according the order to achieve the above, the first detection unit for the detection of radon and its daughter nuclides as well as Tron and its daughter nuclides in the gas to be measured , the first detection unit and the serially coupled, the measurement gas which has been discharged from the first detection unit is introduced, a second detection unit for detecting radon gas to be measured, said first detection a radon thoron measuring apparatus comprising a radon extraction means provided, the between the a unit second detection unit, wherein the first detection unit, a first gas introducing chamber to the measurement gas is introduced When, close to the a first radiation detector arranged so desired a first α-ray incident window to a first gas introducing chamber, wherein the first α-ray incident window in the first gas introducing chamber and a first one-side electrodes disposed, a first other-side electrode in the first gas introducing chamber which is disposed opposite to the first one-side electrode, the first one-side electrode to have a polarity to collect radon daughters and thoron daughters, wherein the first voltage applying means for applying a voltage between the said first of the one-side electrode and the first other-side electrode, said second the detection unit, a second in which the measurement gas has passed through the radon extraction means are arranged so desired a second gas introducing chamber to be introduced, said second second α-ray incident window to the gas introduction chamber a radiation detector, and a second one side electrode disposed in proximity to said second α-ray incident window in the second gas introducing chamber, one of said second in the second gas introducing chamber with the polarity of collecting a second other-side electrode disposed opposite to the side electrode, the radon daughter nuclides generated after passage of the radon extraction unit to the second one side electrode, and the second one side electrode characterized in that it comprises a and a second voltage applying means for applying a voltage between said second other-side electrode.

[**0012**] According to a second aspect of the invention, said radon extraction means, characterized in that it is a filter paper filter having a predetermined thickness.

[**0013**] According to a third aspect of the invention, said radon extraction unit may measurement gas discharged from the first detection unit is a charging substance removing apparatus that is introduced after a predetermined time.

[**0014**] Fourth aspect of the present invention, the signal processing unit, using the output of the output and the second detection unit of the first detection unit, and obtains the radon count and thoron count.

[**0015**] [Action] According to the arrangement according to claim 1, when the measurement gas is introduced into the gas introduction chamber of the first detection unit, the electric field between the first one-side electrode and the first other-side electrode by, radon progeny and thoron daughter nuclide is charged nuclides are captured are attracted to the first of the one-side electrode. On the other hand, Radon and thoron which is the parent nuclide is uniformly dispersed in the first gas introducing chamber.

[**0016**] And, by the first radiation detector, the first of the one-side electrode to the captured radon daughters and α-rays from Tron daughter, and, α-rays from the gas introduced indoor radon and thoron is detected. In this case, it enhanced the detection efficiency since it is possible to collect the daughter nuclides in the vicinity of the detector by an electric field action.

[**0017**] The measurement gas discharged from the first detection unit passes through the radon extraction means, radon progeny, thoron and thoron daughters are removed. Then introduced into the second detection unit.

[**0018**] In the second detection unit, and a daughter nuclide occurs in radon decay in the interior, by the action of an electric field, are captured are attracted to the second one side electrode. Therefore, in the second of the gas introduction chamber, the other of radon, but will also be present its daughter nuclide, for radon progeny are those originally shown the radon in the measurement gas, detection including it by, it enhanced the detection sensitivity of radon itself. That is, in the second of radiation measuring instrument, α-rays from α-ray and radon daughter nuclides from radon is detected.

[**0019**] In short, in the first detection unit, to obtain the measurement results of the total, in the second detection unit are obtained measurement results of radon, it is possible to increase the detection efficiency in any case.

[**0020**] According to the configuration of claim 2, radon (radon gas) is extracted by the filter paper filter. In other words, radon progeny and thoron daughters because of the individual, are blocked by

the filter paper filter, because it is a gas Tron is to transfer to an individual is but a short daughter nuclide has a half-life to enter into the filter paper, it is still blocked It will be.

[**0021**] According to the configuration of claim 3 wherein, prior to being introduced into the charging material removal device, Tron in the measurement gas which has been discharged from the first detection unit is changed to the daughters collapsed in such a pipe. Therefore, if removed radon daughters and thoron daughters by the charging substance removing apparatus, as a result, it is possible to transfer only the radon to the second detection unit.

[**0022**] According to the configuration described in claim 4, the signal processing unit, and is determined separately Radon count and thoron count.

[**0023**] Example will be described below on the basis of a preferred embodiment in the drawings of the present invention.

[**0024**] In Figure 1, the overall configuration of the radon thoron measuring apparatus according to the present invention is shown. This radon Tron measuring device, roughly, and a detection unit 24 and the signal processing unit 26. Detecting section 24 is formed by the first detection unit 30 and the second detection unit 32 connected in cascade.

[**0025**] Configuration of the detector 24 is shown in detail in FIG. In the first detection unit 30, the measurement vessel 34 to form a gas introduction chamber which is constituted by an insulating member, the ionization chamber 36 is disposed. The α-ray incident window 44 of ionization chamber 36 is formed so as to constitute a part of the measuring container 34 wants the measuring container 34. α-ray incident window 44 is extremely thin α-ray transmission member, composed of, for example, mica. Then, the conductive coating is applied on its inner surface.

[**0026**] Shell 37 of the ionization chamber 36 is formed of a metal, the outer electrode composed of a shell 37 and the conductive coating. That is, the interior of the ionization chamber 36, two insulating members 38 and the linear collector electrode 42 which is supported by 40 is arranged in parallel with the α-ray incident window 44, between the collector electrode and the outer electrode high voltage (for example, 500V) is applied to. In addition, the voltage application is performed so that the outer electrode is a negative electrode.

[**0027**] In the measuring container 34, one side electrode 46 of the adjacent thereto parallel to the planar to the α-ray incident window 44 is disposed while being insulated from other members. In addition, it opposed the other side electrode 48 parallel to the surface of the spaced a certain distance has been placed.

[0028] Between the electrodes 46 and 48, a high voltage (for example, 1000~2000V) is applied by the power supply 50. In that case, the electrode 46 is at the negative electrode, the electrode 48 is the positive electrode. In this embodiment, the distance between the electrodes 46 and 48, considering that the projected range of α rays is about approximately 5 cm, there is a 3 cm.

[0029] The second detection unit 32 has the same configuration as the first detection unit 30, the components identical to members of the first detection unit 30 denoted by reference numerals obtained by adding 100 number on the sign, its description omitted.

[0030] 2, the front of the first detection unit 30, the dust filter 52 is disposed. Radon extraction means 54 between the first detection unit 30 and the second detection unit 32, a non-measurement gas is a means to extract only radon gas, in other words, radon daughters, a thoron and thoron daughters removed it is a device that. For example, it is possible to use a filter paper filter having a predetermined thickness. That is, according to the filter paper filter, for radon progeny and thoron daughter nuclide is blocked because of the individual, is a gas Tron is to transfer to an individual is but a short daughter nuclide has a half-life to enter into the filter paper, still blocking It is. Therefore, the thickness of the gas transit time in the filter paper is determined to be larger than the half-life of thoron, the filter paper is used having a thickness, for example with a several minutes to 10 minutes about gas transit time .

[0031] Further, as the radon extraction unit 54, it can be utilized charged substance removing apparatus such as ion emission pre-recipe theta. In that case, between the charging material removal device and the first detection unit 30, it is arranged a pipe 200 with a length that almost every Tron disintegrate in movement. In other words, the half-life of Tron is relatively short, after the introduction of the detection unit 24, since most of Tron in the middle of moving After a lapse of more than a few minutes to 10 minutes to collapse, the Tron as a result if long a pipe route it is possible to remove, if Okurikome such Tron gas to be measured after the collapse to the charging material removal device, it is removed radon daughters and thoron daughters, only finally radon the second detection unit 32 it can be fed in.

[0032] The subsequent stage of the second detection unit 32, a pump 56 for forming a flow of the measurement gas near the natural convection is disposed.

[0033] Next is a description of the operation and the signal processing unit 26 of the detector 24.

[0034] 2, is introduced into the measurement gas is first the first detection unit 30. Then, radon and thoron is evenly distributed, and their daughter is captured drawn to one side electrode 46. And, on the other hand trapped in the side electrode 46 the radon progeny and α-rays from Tron daughter, and, in the ionization chamber 36 through the α line is α-ray incident window 44 from the radon

and thoron in the measuring container 34 It is detected on. Specifically, in the ionization chamber 36 is sealed air or ionized gas at 1 atm as ionized gas, alpha rays to ionize the gas, thereby resulting electrons are captured by the collector electrode 42. This detection pulse is output.

[**0035**] The measurement gas which has been discharged from the first detection unit 30, via the radon extraction unit 54 described above, radon progeny, thoron and thoron daughters are removed, the measurement gas and the second detection unit containing only radon 32 to reach.

[**0036**] In the second detection unit 32, the other alpha-rays from radon, alpha rays from radon daughters caused by collapse after passing through the radon extraction unit 54 is detected. In this case, in the vicinity of the α-ray incident window 144, one side electrode 146 is disposed, since radon daughters are collected are attracted to one side electrode 146, the detection sensitivity of radon daughters in turn radon itself It is enhanced. Is the same as that of the ionization chamber 36 for detecting the action of α-rays due to the ionization chamber 136.

[**0037**] 1, the detection pulse outputted from the first detection unit is amplified by a preamplifier 58 and amplifier 60, and passes through the further not shown the waveform shaping circuit and the pulse-height discriminator circuit is counted by the counting circuit 62 . The count result will indicate the count value of radon and its daughter nuclides as well as Tron and its daughter nuclide total.

[**0038**] On the other hand, the delay coincidence circuit 64 is a circuit for obtaining a count of only Tron. Half-life of 216Po resulting from Tron is very short as 0.15 seconds, so that the continuous detection pulse of immediately 216Po subsequent to the detection pulse of this for Tron. Therefore, those utilizing the phenomenon, a certain period of time (e.g., one second) detecting when two pulses are continuous in the (detection of the continuous pulse), to estimate the detection Tron if it is detected it is. Of course, the probabilistic Tron - but can be detected even if continuous two pulses outside 216Po, the concentration of such radon monitor environmental radiation its probability for low low, less influence on measurement accuracy. Thus, the output of the delay coincidence circuit 64 indicates the count value of only Tron.

[**0039**] On the other hand, the detection pulse from the second detection unit 32 is amplified by a preamplifier 68 and amplifier 70, and passes through the further not shown the waveform shaping circuit and the pulse-height discriminator circuit is counted by the counting circuit 72. The count result will exhibit a count value of radon.

[**0040**] Then, the adder circuit 66, it is possible to obtain an output of the delay coincidence circuit 64, when adding the output of the counting circuit 72, the count value of the parent (radon count + TRON count value), the difference arithmetic circuit 74 in, the output of the counting circuit 62, an output of the delay coincidence circuit 64, if the subtraction output of the counting circuit 72,

and is possible to obtain the count of the daughter nuclide (RN count + thoron daughters count) it can. Incidentally, when each counting and calculation, performs correction in consideration of the balance of the detection efficiency as needed.

[**0041**] Thus, according to the signal processing section 26, it is possible to find up to the breakdown in detail including the count value of the total.

[**0042**] In Figure 3, specific examples of electrode 46,48,146,148 is shown. Examples of this electrode, (A), the as shown in can be formed extremely thin conductive film shape, or (Bs), can also be a parallel wire structures. Or, it may be a lattice thin line structure shown in (C). The electrode 46 and 146 are radionuclides for accumulating increases with time, there is a replaceable to adhere to it.

[**0043**] As described above, according to the present invention, according to the present invention, the radon and thoron can each measured separately. In addition, it is possible to realize a measurement of radon progeny and thoron daughter nuclides. Furthermore, to improve the measurement sensitivity, real-time, simple, economically measurements.

[**0044**] Brief description of the drawings

[**0045**] Is a diagram showing an entire configuration of a radiation measuring device according to FIG 1 the present invention.

[**0046**] It is a diagram showing a specific structure of FIG detector.

[**0047**] Is a diagram illustrating an example of the structure of FIG. 3 electrodes.

[**0048**] Figure 4 is a diagram showing a configuration of a conventional radon thoron monitor.

[**0049**] It is a diagram illustrating a FIG. 5 radon-series and thoron-series.

[**0050**] Description of the code

[**0051**] 24 detection section 26 signal processing unit 30 first detecting unit 32 and the second detection unit 54 radon extracting means

A first detection unit for detecting radon and its daughters and thoron and their daughter nuclides in the measurement gas, is connected to the first detection unit and the series, measured discharged from the first detection unit gas is introduced, and a second detection unit for detecting radon in a measurement

gas, radon thoron containing a radon extraction means provided between said first detection unit and the second detection unit a measuring device, wherein the first detection unit is disposed so desired the first gas introducing chamber to the measurement gas is introduced, said first first α-ray incident window to the gas introduction chamber a first radiation detector, a first one side electrode disposed in proximity to the first α-ray incident window in the first gas introducing chamber, the first in said first gas introducing chamber a first other-side electrode disposed opposite to one side electrode of said to first one side electrode with a polarity to collect radon daughters and thoron daughters, said first and said first one-side electrode a first voltage applying means for applying a voltage between the other side electrodes, said second detection unit, a second gas introducing chamber to the measurement gas that has passed through the radon extraction means is introduced, the a second radiation detector arranged so desired a second α-ray incident window to the second gas introducing chamber, arranged close to each other in the second gas introducing chamber to said second α-ray incident window a second one side electrodes, and a second other-side electrode disposed to face the second one side electrode in the second gas introducing chamber, generated after passage of the radon extraction means with a polarity to collect radon daughters in the second one side electrode, in that it comprises a second voltage applying means for applying a voltage between the said and the second one side electrode and the second on the other side electrode radon Tron measuring device which is characterized.

The apparatus of claim 1, wherein the radon extraction means, radon thoron measuring device, characterized in that the filter paper filter having a predetermined thickness.

The apparatus of claim 1, wherein the radon extraction means, radon thoron measuring device, characterized in that the measurement gas which has been discharged from the first detection unit is a charging substance removing apparatus that is introduced after a predetermined time.

The apparatus of claim 1, wherein the signal processing unit, the first using the output of the output and the second detection unit of the detection unit, radon Tron, characterized in that for determining the radon count and thoron count measuring device.